Individual Project Check-In 1

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Writing Note

This check-in is based on my direct classroom observations in AP Computer Science and my own low-fidelity design drafts and study plan. The example discussed in this document (e.g., counter pattern with reset mechanism) reflects common mistakes I have observed from my students in FRQ practice.

1 PROJECT OVERVIEW

Background & Motivation

In AP CS, free-response questions are more challenging than multiple-choice questions. Most students have no idea how to start or get stuck at a certain step because FRQ questions require them to write out their algorithm in a clear, structured way. In this case, timely and valid feedback is essential for improvement.

In my observation, students in AP classes tend to focus on scores: they rarely review the feedback report, or they don't know how to improve for the next attempt, so they heavily rely on their teacher's explanations. For instance, many questions test the counter pattern combined with a reset mechanism. Many students don't know when or how to reset the counter variable to zero, resulting in a loss of points on FRQs.

Thus, I hope to redesign the FRQ score report not only to explain why students lose points, but also to guide them on which skills to practice, which questions test those skills, and how to improve their performance on similar questions. Ultimately, I want to translate the score report into a practical improvement plan.

What I am redesigning:

What I am redesigning is the AP Classroom AP CS FRQ student-side score report interface, where students can see how many points they get and the corresponding scoring.

Why it matters:

Valid feedback can greatly reduce reliance on teachers' explanations by turning the reasons for lost points into guidance for next steps. For FRQs that require consistent reflection and targeted practice, if feedback is not detailed and students only focus on their score, their confidence and willingness to practice can be compromised. Designing the feedback interface to be more actionable can equip students to improve their performance on similar questions in a more independent and effective way.

2 DESIGN EXPLORATION (THREE ALTERNATIVES)

Design A: Rubric Breakdown

Design intent:

This design highlights transparency and structure. The goal is to enable students to see how many points they earned for each scoring criterion, so they can understand where points are lost.

Key elements:

- Display each scoring criterion
- Explanation for each scoring criterion
- Highlight where points are lost
- Organized information layout

Design B: Anchored Feedback

Design intent:

This design emphasizes a clear mapping between students' answers and the scoring rubric, reducing the cognitive load of understanding why points are lost.

Key elements:

- Side-by-side view of students' answers and comments
- Clear mapping between each comment and the relevant part of the student's answer
- Expandable comments that reveal detailed explanations on click/tap
- A consolidated summary of common issue types (frequent mistakes)

Design C: Practice Plan

Design intent:

This design emphasizes translating feedback into an actionable practice plan.

Key elements:

- Categorization of common mistake types
- Skill tags aligned with each mistake
- Recommended practice directions or similar questions
- A short "next-step practice plan" section

Low-Fidelity Wireframes

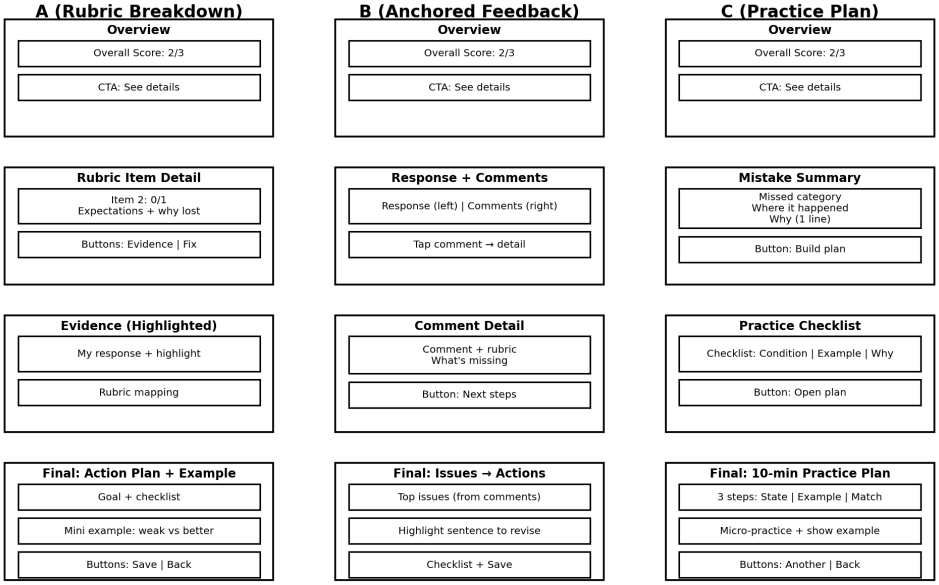

Figure 1—Low-fidelity wireframes for the three alternatives (A/B/C).

3 QUESTIONS FOR PEER FEEDBACK

1. Do the three designs meaningfully differ, or are they too similar?
2. Is my framing of the student goal (find → understand → act) appropriate for FRQ feedback?
3. Which design direction seems most promising to iterate on, and why?
4. What key information or workflow step is missing from all three?

5. Any concerns about scope (too broad/too narrow) for the full project?